Exploratory Data Analysis

Agenda

In this session, we will cover the following concepts with the help of a business use case:

- Discover patterns in the dataset
- Spot anomalies in the dataset
- Frame hypothesis from the dataset
- Gain insights by plotting different kinds of graphs using Python libraries, Seaborn, and Matplotlib

Now let's see the above concepts with help of a use case

Problem Statement:

There are so many variables that impact the price of a house. With dynamic parameters in the residential real state business, it is always important to reach a reasonable price for better business opportunities. As a part of the analytics team in a real state company, you have to come up with the variables that are impacting the price of the house through analyzing and visualizing the data.

Dataset

Download the dataset "housing_data.csv" from Course Resources section and upload the dataset in the lab.

Data Dictionary

MSSubClass: Identifies the type of dwelling involved in the sale.

```
 20  1-STORY 1946 & NEWER ALL STYLES
 30  1-STORY 1945 & OLDER
 40  1-STORY W/FINISHED ATTIC ALL AGES
 45  1-1/2 STORY - UNFINISHED ALL AGES
 50  1-1/2 STORY FINISHED ALL AGES
 60  2-STORY 1946 & NEWER
 70  2-STORY 1945 & OLDER
 75  2-1/2 STORY ALL AGES
 80  SPLIT OR MULTI-LEVEL
 85  SPLIT FOYER
 90  DUPLEX - ALL STYLES AND AGES
120  1-STORY PUD (Planned Unit Development) - 1946 & NEWER
150  1-1/2 STORY PUD - ALL AGES
160  2-STORY PUD - 1946 & NEWER
180  PUD - MULTILEVEL - INCL SPLIT LEV/FOYER
190  2 FAMILY CONVERSION - ALL STYLES AND AGES
```

MSZoning: Identifies the general zoning classification of the sale.

```
A    Agriculture
C    Commercial
FV   Floating Village Residential
I    Industrial
RH   Residential High Density
RL   Residential Low Density
RP   Residential Low Density Park
RM   Residential Medium Density
```

LotFrontage: Linear feet of street connected to property

LotArea: Lot size in square feet

Street: Type of road access to property

```
Grvl Gravel
Pave Paved
```

Alley: Type of alley access to property

```
Grvl Gravel
Pave Paved
NA   No alley access
```

LotShape: General shape of property

```
Reg  Regular
IR1  Slightly irregular
IR2  Moderately Irregular
IR3  Irregular
```

LandContour: Flatness of the property

```
Lvl  Near Flat/Level
Bnk  Banked - Quick and significant rise from street grade to building
HLS  Hillside - Significant slope from side to side
Low  Depression
```

Utilities: Type of utilities available

```
AllPub   All public Utilities (E,G,W,& S)
NoSewr   Electricity, Gas, and Water (Septic Tank)
NoSeWa   Electricity and Gas Only
ELO  Electricity only
```

LotConfig: Lot configuration

```
Inside   Inside lot
Corner   Corner lot
CulDSac  Cul-de-sac
FR2  Frontage on 2 sides of property
FR3  Frontage on 3 sides of property
```

LandSlope: Slope of property

```
Gtl  Gentle slope
Mod  Moderate Slope
Sev  Severe Slope
```

Neighborhood: Physical locations within Ames city limits

```
Blmngtn  Bloomington Heights
Blueste  Bluestem
BrDale   Briardale
BrkSide  Brookside
ClearCr  Clear Creek
CollgCr  College Creek
Crawfor  Crawford
Edwards  Edwards
Gilbert  Gilbert
IDOTRR   Iowa DOT and Rail Road
MeadowV  Meadow Village
Mitchel  Mitchell
Names    North Ames
NoRidge  Northridge
NPkVill  Northpark Villa
NridgHt  Northridge Heights
NWAmes   Northwest Ames
OldTown  Old Town
SWISU    South & West of Iowa State University
Sawyer   Sawyer
SawyerW  Sawyer West
Somerst  Somerset
StoneBr  Stone Brook
Timber   Timberland
Veenker  Veenker
```

Condition1: Proximity to various conditions

```
Artery   Adjacent to arterial street
Feedr    Adjacent to feeder street
Norm Normal
RRNn Within 200' of North-South Railroad
RRAn Adjacent to North-South Railroad
PosN Near positive off-site feature--park, greenbelt, etc.
PosA Adjacent to postive off-site feature
RRNe Within 200' of East-West Railroad
RRAe Adjacent to East-West Railroad
```

Condition2: Proximity to various conditions (if more than one is present)

```
Artery   Adjacent to arterial street
Feedr    Adjacent to feeder street
Norm Normal
RRNn Within 200' of North-South Railroad
RRAn Adjacent to North-South Railroad
PosN Near positive off-site feature--park, greenbelt, etc.
PosA Adjacent to postive off-site feature
RRNe Within 200' of East-West Railroad
RRAe Adjacent to East-West Railroad
```

BldgType: Type of dwelling

```
1Fam Single-family Detached
2FmCon   Two-family Conversion; originally built as one-family dwelling
Duplx    Duplex
TwnhsE   Townhouse End Unit
TwnhsI   Townhouse Inside Unit
```

HouseStyle: Style of dwelling

```
1Story   One story
1.5Fin   One and one-half story: 2nd level finished
1.5Unf   One and one-half story: 2nd level unfinished
2Story   Two story
2.5Fin   Two and one-half story: 2nd level finished
2.5Unf   Two and one-half story: 2nd level unfinished
SFoyer   Split Foyer
SLvl Split Level
```

OverallQual: Rates the overall material and finish of the house

```
10   Very Excellent
9    Excellent
8    Very Good
7    Good
6    Above Average
5    Average
4    Below Average
3    Fair
2    Poor
1    Very Poor
```

OverallCond: Rates the overall condition of the house

```
10   Very Excellent
9    Excellent
8    Very Good
7    Good
6    Above Average
5    Average
4    Below Average
3    Fair
2    Poor
1    Very Poor
```

YearBuilt: Original construction date

YearRemodAdd: Remodel date (same as construction date if no remodeling or additions)

RoofStyle: Type of roof

```
Flat Flat
Gable    Gable
Gambrel  Gabrel (Barn)
Hip  Hip
Mansard  Mansard
Shed Shed
```

RoofMatl: Roof material

```
ClyTile  Clay or Tile
CompShg  Standard (Composite) Shingle
Membran  Membrane
Metal    Metal
Roll Roll
Tar&Grv  Gravel & Tar
WdShake  Wood Shakes
WdShngl  Wood Shingles
```

Exterior1st: Exterior covering on house

```
AsbShng  Asbestos Shingles
AsphShn  Asphalt Shingles
BrkComm  Brick Common
BrkFace  Brick Face
CBlock   Cinder Block
CemntBd  Cement Board
HdBoard  Hard Board
ImStucc  Imitation Stucco
MetalSd  Metal Siding
Other    Other
Plywood  Plywood
PreCast  PreCast
Stone    Stone
Stucco   Stucco
VinylSd  Vinyl Siding
Wd Sdng  Wood Siding
WdShing  Wood Shingles
```

Exterior2nd: Exterior covering on house (if more than one material)

```
AsbShng  Asbestos Shingles
AsphShn  Asphalt Shingles
BrkComm  Brick Common
BrkFace  Brick Face
CBlock   Cinder Block
CemntBd  Cement Board
HdBoard  Hard Board
ImStucc  Imitation Stucco
MetalSd  Metal Siding
Other    Other
Plywood  Plywood
PreCast  PreCast
Stone    Stone
Stucco   Stucco
VinylSd  Vinyl Siding
Wd Sdng  Wood Siding
WdShing  Wood Shingles
```

MasVnrType: Masonry veneer type

```
BrkCmn   Brick Common
BrkFace  Brick Face
CBlock   Cinder Block
None None
Stone    Stone
```

MasVnrArea: Masonry veneer area in square feet

ExterQual: Evaluates the quality of the material on the exterior

```
Ex   Excellent
Gd   Good
TA   Average/Typical
Fa   Fair
Po   Poor
```

ExterCond: Evaluates the present condition of the material on the exterior

```
Ex   Excellent
Gd   Good
TA   Average/Typical
Fa   Fair
Po   Poor
```

Foundation: Type of foundation

```
BrkTil   Brick & Tile
CBlock   Cinder Block
PConc    Poured Contrete
Slab Slab
Stone    Stone
Wood Wood
```

BsmtQual: Evaluates the height of the basement

```
Ex   Excellent (100+ inches)
Gd   Good (90-99 inches)
TA   Typical (80-89 inches)
Fa   Fair (70-79 inches)
Po   Poor (<70 inches
NA   No Basement
```

BsmtCond: Evaluates the general condition of the basement

```
Ex   Excellent
Gd   Good
TA   Typical - slight dampness allowed
Fa   Fair - dampness or some cracking or settling
Po   Poor - Severe cracking, settling, or wetness
NA   No Basement
```

BsmtExposure: Refers to walkout or garden level walls

```
Gd   Good Exposure
Av   Average Exposure (split levels or foyers typically score average or above)
Mn   Mimimum Exposure
No   No Exposure
NA   No Basement
```

BsmtFinType1: Rating of basement finished area

```
GLQ  Good Living Quarters
ALQ  Average Living Quarters
BLQ  Below Average Living Quarters
Rec  Average Rec Room
LwQ  Low Quality
Unf  Unfinshed
NA   No Basement
```

BsmtFinSF1: Type 1 finished square feet

BsmtFinType2: Rating of basement finished area (if multiple types)

```
GLQ  Good Living Quarters
ALQ  Average Living Quarters
BLQ  Below Average Living Quarters
Rec  Average Rec Room
LwQ  Low Quality
Unf  Unfinshed
NA   No Basement
```

BsmtFinSF2: Type 2 finished square feet

BsmtUnfSF: Unfinished square feet of basement area

TotalBsmtSF: Total square feet of basement area

Heating: Type of heating

```
Floor    Floor Furnace
GasA Gas forced warm air furnace
GasW Gas hot water or steam heat
Grav Gravity furnace
OthW Hot water or steam heat other than gas
Wall Wall furnace
```

HeatingQC: Heating quality and condition

```
Ex   Excellent
Gd   Good
TA   Average/Typical
Fa   Fair
Po   Poor
```

CentralAir: Central air conditioning

```
N    No
Y    Yes
```

Electrical: Electrical system

```
SBrkr    Standard Circuit Breakers & Romex
FuseA    Fuse Box over 60 AMP and all Romex wiring (Average)
FuseF    60 AMP Fuse Box and mostly Romex wiring (Fair)
FuseP    60 AMP Fuse Box and mostly knob & tube wiring (poor)
Mix  Mixed
```

1stFlrSF: First Floor square feet

2ndFlrSF: Second floor square feet

LowQualFinSF: Low quality finished square feet (all floors)

GrLivArea: Above grade (ground) living area square feet

BsmtFullBath: Basement full bathrooms

BsmtHalfBath: Basement half bathrooms

FullBath: Full bathrooms above grade

HalfBath: Half baths above grade

Bedroom: Bedrooms above grade (does NOT include basement bedrooms)

Kitchen: Kitchens above grade

KitchenQual: Kitchen quality

```
Ex   Excellent
Gd   Good
TA   Typical/Average
Fa   Fair
Po   Poor
```

TotRmsAbvGrd: Total rooms above grade (does not include bathrooms)

Functional: Home functionality (Assume typical unless deductions are warranted)

```
Typ  Typical Functionality
Min1 Minor Deductions 1
Min2 Minor Deductions 2
Mod  Moderate Deductions
Maj1 Major Deductions 1
Maj2 Major Deductions 2
Sev  Severely Damaged
Sal  Salvage only
```

Fireplaces: Number of fireplaces

FireplaceQu: Fireplace quality

```
Ex   Excellent - Exceptional Masonry Fireplace
Gd   Good - Masonry Fireplace in main level
TA   Average - Prefabricated Fireplace in main living area or Masonry Fireplace in basement
Fa   Fair - Prefabricated Fireplace in basement
Po   Poor - Ben Franklin Stove
NA   No Fireplace
```

GarageType: Garage location

```
2Types   More than one type of garage
Attchd   Attached to home
Basment  Basement Garage
BuiltIn  Built-In (Garage part of house - typically has room above garage)
CarPort  Car Port
Detchd   Detached from home
NA   No Garage
```

GarageYrBlt: Year garage was built

GarageFinish: Interior finish of the garage

```
Fin  Finished
RFn  Rough Finished
Unf  Unfinished
NA   No Garage
```

GarageCars: Size of garage in car capacity

GarageArea: Size of garage in square feet

GarageQual: Garage quality

```
Ex   Excellent
Gd   Good
TA   Typical/Average
Fa   Fair
Po   Poor
NA   No Garage
```

GarageCond: Garage condition

```
Ex   Excellent
Gd   Good
TA   Typical/Average
Fa   Fair
Po   Poor
NA   No Garage
```

PavedDrive: Paved driveway

```
Y    Paved
P    Partial Pavement
N    Dirt/Gravel
```

WoodDeckSF: Wood deck area in square feet

OpenPorchSF: Open porch area in square feet

EnclosedPorch: Enclosed porch area in square feet

3SsnPorch: Three season porch area in square feet

ScreenPorch: Screen porch area in square feet

PoolArea: Pool area in square feet

PoolQC: Pool quality

```
Ex   Excellent
Gd   Good
TA   Average/Typical
Fa   Fair
NA   No Pool
```

Fence: Fence quality

```
GdPrv    Good Privacy
MnPrv    Minimum Privacy
GdWo Good Wood
MnWw Minimum Wood/Wire
NA   No Fence
```

MiscFeature: Miscellaneous feature not covered in other categories

```
Elev Elevator
Gar2 2nd Garage (if not described in garage section)
Othr Other
Shed Shed (over 100 SF)
TenC Tennis Court
NA   None
```

MiscVal: $Value of miscellaneous feature

MoSold: Month Sold (MM)

YrSold: Year Sold (YYYY)

SaleType: Type of sale

```
WD   Warranty Deed - Conventional
CWD  Warranty Deed - Cash
VWD  Warranty Deed - VA Loan
New  Home just constructed and sold
COD  Court Officer Deed/Estate
Con  Contract 15% Down payment regular terms
ConLw    Contract Low Down payment and low interest
ConLI    Contract Low Interest
ConLD    Contract Low Down
Oth  Other
```

SaleCondition: Condition of sale

```
Normal   Normal Sale
Abnorml  Abnormal Sale -  trade, foreclosure, short sale
AdjLand  Adjoining Land Purchase
Alloca   Allocation - two linked properties with separate deeds, typically condo with a garage unit
Family   Sale between family members
Partial  Home was not completed when last assessed (associated with New Homes)
```

Initial Exploration

Learning the basics of exploratory data analysis using Python with Numpy, Matplotlib, and Pandas.

What is EDA?

In Python, EDA is used for data visualization to generate meaningful patterns and insights from data. It is also used for cleaning the data before analyzing it to remove any abnormalities.

Based on the results of EDA, companies also make business decisions, such as:

1- If EDA isn't done correctly, it can affect the further steps used in model building.

2- If EDA is done properly, it can improve the efficiency of any model.

Import necessary packages

In Python, Numpy is a package that includes multidimensional array objects as well as a number of derived objects.

Matplotlib is an amazing visualization library in Python for 2D plots of arrays.

Pandas are used for data manipulation and analysis. So these are the core libraries that are used for the EDA process.

These libraries are written with an import keyword.

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

In [2]:
```
%matplotlib inline
```

Let's read the data

Observations:

- We have to upload the dataset in the file explorer on the left panel of your lab
- We are reading the file through the dataframe variable
- The file is in CSV format
- We use the `pd.read_csv()` function to read a CSV file
- We provide the exact path of the file within the round bracket `()`

In [3]:
```
dataframe = pd.read_csv('housing_data.csv', index_col=0)
```

pd.read_csv function is used to read the "housing_data.csv" file. If index_col=0, the function will take the first column as an index by default, whereas if index_col=None, then the first row is not used as the column name.

dataframe is a variable which will store the data read by the csv file.

Print first 5 lines of the dataset

In [4]:
```
dataframe.head(5)
```

Out[4]:

	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour	Utilities	LotConfig	...	PoolArea	PoolQC	Fence	M
0	SC60	RL	65.0	8450	Pave	None	Reg	Lvl	AllPub	Inside	...	0	No	No	
1	SC20	RL	80.0	9600	Pave	None	Reg	Lvl	AllPub	FR2	...	0	No	No	
2	SC60	RL	68.0	11250	Pave	None	IR1	Lvl	AllPub	Inside	...	0	No	No	
3	SC70	RL	60.0	9550	Pave	None	IR1	Lvl	AllPub	Corner	...	0	No	No	
4	SC60	RL	84.0	14260	Pave	None	IR1	Lvl	AllPub	FR2	...	0	No	No	

5 rows × 80 columns

head(5) function prints the first five rows from the dataframe.

Print last 5 lines of the dataset

tail(n=5) prints the last five rows from the dataframe.

In [5]:
```
dataframe.tail(5)
```

Out[5]:

	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour	Utilities	LotConfig	...	PoolArea	PoolQC	Fence
1455	SC60	RL	62.0	7917	Pave	None	Reg	Lvl	AllPub	Inside	...	0	No	No
1456	SC20	RL	85.0	13175	Pave	None	Reg	Lvl	AllPub	Inside	...	0	No	MnPrv
1457	SC70	RL	66.0	9042	Pave	None	Reg	Lvl	AllPub	Inside	...	0	No	GdPrv
1458	SC20	RL	68.0	9717	Pave	None	Reg	Lvl	AllPub	Inside	...	0	No	No
1459	SC20	RL	75.0	9937	Pave	None	Reg	Lvl	AllPub	Inside	...	0	No	No

5 rows × 80 columns

Check the size of the dataset

Shape is an attribute that returns a tuple representing the dimensionality of the dataframe. Shape is used to define the number of rows and columns in a dataframe.

In [6]:
```
dataframe.shape
```

Out[6]:
```
(1460, 80)
```

It is also good practice to know the columns and their corresponding data types, along with finding whether they contain null values or not. For this, info() method is used to print the information about the dataframe.

In [7]:
```
dataframe.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 1460 entries, 0 to 1459
Data columns (total 80 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   MSSubClass     1460 non-null   object
 1   MSZoning       1460 non-null   object
 2   LotFrontage    1460 non-null   float64
 3   LotArea        1460 non-null   int64
 4   Street         1460 non-null   object
 5   Alley          1460 non-null   object
 6   LotShape       1460 non-null   object
 7   LandContour    1460 non-null   object
 8   Utilities      1460 non-null   object
 9   LotConfig      1460 non-null   object
 10  LandSlope      1460 non-null   object
 11  Neighborhood   1460 non-null   object
 12  Condition1     1460 non-null   object
 13  Condition2     1460 non-null   object
 14  BldgType       1460 non-null   object
 15  HouseStyle     1460 non-null   object
 16  OverallQual    1460 non-null   int64
 17  OverallCond    1460 non-null   int64
 18  YearBuilt      1460 non-null   int64
 19  YearRemodAdd   1460 non-null   int64
 20  RoofStyle      1460 non-null   object
 21  RoofMatl       1460 non-null   object
 22  Exterior1st    1460 non-null   object
 23  Exterior2nd    1460 non-null   object
 24  MasVnrType     1460 non-null   object
 25  MasVnrArea     1460 non-null   float64
 26  ExterQual      1460 non-null   object
 27  ExterCond      1460 non-null   object
 28  Foundation     1460 non-null   object
 29  BsmtQual       1460 non-null   object
 30  BsmtCond       1460 non-null   object
 31  BsmtExposure   1460 non-null   object
 32  BsmtFinType1   1460 non-null   object
 33  BsmtFinSF1     1460 non-null   int64
 34  BsmtFinType2   1460 non-null   object
 35  BsmtFinSF2     1460 non-null   int64
 36  BsmtUnfSF      1460 non-null   int64
 37  TotalBsmtSF    1460 non-null   int64
 38  Heating        1460 non-null   object
 39  HeatingQC      1460 non-null   object
 40  CentralAir     1460 non-null   object
 41  Electrical     1459 non-null   object
 42  1stFlrSF       1460 non-null   int64
 43  2ndFlrSF       1460 non-null   int64
 44  LowQualFinSF   1460 non-null   int64
 45  GrLivArea      1460 non-null   int64
 46  BsmtFullBath   1460 non-null   int64
 47  BsmtHalfBath   1460 non-null   int64
 48  FullBath       1460 non-null   int64
 49  HalfBath       1460 non-null   int64
 50  BedroomAbvGr   1460 non-null   int64
 51  KitchenAbvGr   1460 non-null   int64
 52  KitchenQual    1460 non-null   object
 53  TotRmsAbvGrd   1460 non-null   int64
 54  Functional     1460 non-null   object
 55  Fireplaces     1460 non-null   int64
 56  FireplaceQu    1460 non-null   object
 57  GarageType     1460 non-null   object
 58  GarageYrBlt    1379 non-null   float64
 59  GarageFinish   1460 non-null   object
 60  GarageCars     1460 non-null   int64
 61  GarageArea     1460 non-null   int64
 62  GarageQual     1460 non-null   object
 63  GarageCond     1460 non-null   object
 64  PavedDrive     1460 non-null   object
 65  WoodDeckSF     1460 non-null   int64
 66  OpenPorchSF    1460 non-null   int64
 67  EnclosedPorch  1460 non-null   int64
 68  3SsnPorch      1460 non-null   int64
 69  ScreenPorch    1460 non-null   int64
 70  PoolArea       1460 non-null   int64
 71  PoolQC         1460 non-null   object
 72  Fence          1460 non-null   object
 73  MiscFeature    1460 non-null   object
 74  MiscVal        1460 non-null   int64
 75  MoSold         1460 non-null   object
 76  YrSold         1460 non-null   int64
 77  SaleType       1460 non-null   object
 78  SaleCondition  1460 non-null   object
 79  SalePrice      1460 non-null   int64
dtypes: float64(3), int64(32), object(45)
memory usage: 923.9+ KB
```

Observation: ** There are 80 columns in the dataframe

The `describe()` function helps us to get various summary statistics that exclude NaN values.

This function returns the **count, mean, standard deviation, minimum, and maximum** values as well as the quantiles of the data.

In [8]:

```
dataframe.describe()
```

Out[8]:

	LotFrontage	LotArea	OverallQual	OverallCond	YearBuilt	YearRemodAdd	MasVnrArea	BsmtFinSF1	BsmtFinSF2	BsmtUnfSF
count	1460.000000	1460.000000	1460.000000	1460.000000	1460.000000	1460.000000	1460.000000	1460.000000	1460.000000	1460.00000
mean	57.623288	10516.828082	6.099315	5.575342	1971.267808	1984.865753	103.117123	443.639726	46.549315	567.240411
std	34.664304	9981.264932	1.382997	1.112799	30.202904	20.645407	180.731373	456.098091	161.319273	441.866955
min	0.000000	1300.000000	1.000000	1.000000	1872.000000	1950.000000	0.000000	0.000000	0.000000	0.00000
25%	42.000000	7553.500000	5.000000	5.000000	1954.000000	1967.000000	0.000000	0.000000	0.000000	223.00000
50%	63.000000	9478.500000	6.000000	5.000000	1973.000000	1994.000000	0.000000	383.500000	0.000000	477.50000
75%	79.000000	11601.500000	7.000000	6.000000	2000.000000	2004.000000	164.250000	712.250000	0.000000	808.00000
max	313.000000	215245.000000	10.000000	9.000000	2010.000000	2010.000000	1600.000000	5644.000000	1474.000000	2336.00000

8 rows × 35 columns

Let's find out the number of numerical features in our dataset.

In [9]:

```
numerical_feature_columns = list(dataframe._get_numeric_data().columns)
numerical_feature_columns
```

Out[9]:

```
['LotFrontage',
 'LotArea',
 'OverallQual',
 'OverallCond',
 'YearBuilt',
 'YearRemodAdd',
 'MasVnrArea',
 'BsmtFinSF1',
 'BsmtFinSF2',
 'BsmtUnfSF',
 'TotalBsmtSF',
 '1stFlrSF',
 '2ndFlrSF',
 'LowQualFinSF',
 'GrLivArea',
 'BsmtFullBath',
 'BsmtHalfBath',
 'FullBath',
 'HalfBath',
 'BedroomAbvGr',
 'KitchenAbvGr',
 'TotRmsAbvGrd',
 'Fireplaces',
 'GarageYrBlt',
 'GarageCars',
 'GarageArea',
 'WoodDeckSF',
 'OpenPorchSF',
 'EnclosedPorch',
 '3SsnPorch',
 'ScreenPorch',
 'PoolArea',
 'MiscVal',
 'YrSold',
 'SalePrice']
```

Let's find out the number of categorical features in our dataset.

In [10]:

```
categorical_feature_columns = list(set(dataframe.columns) - set(dataframe._get_numeric_data().columns))
categorical_feature_columns
```

Out[10]:

```
['Neighborhood',
 'BsmtCond',
 'GarageQual',
 'MoSold',
 'BsmtQual',
 'PoolQC',
 'Condition1',
 'BsmtFinType1',
 'Heating',
 'Exterior2nd',
 'GarageType',
 'SaleType',
 'MSSubClass',
 'BsmtExposure',
 'Exterior1st',
 'MSZoning',
 'Condition2',
 'RoofMatl',
 'LotConfig',
 'Utilities',
 'HouseStyle',
 'GarageFinish',
 'GarageCond',
 'ExterQual',
 'FireplaceQu',
 'LandContour',
 'Electrical',
 'HeatingQC',
 'ExterCond',
 'BsmtFinType2',
 'CentralAir',
 'KitchenQual',
 'SaleCondition',
 'LandSlope',
 'MiscFeature',
 'LotShape',
 'PavedDrive',
 'MasVnrType',
 'Functional',
 'Foundation',
 'Alley',
 'BldgType',
 'Street',
 'Fence',
 'RoofStyle']
```

Now, we want to analyze the data in order to extract some insights

There are two major types to analyze the data

- Univariate analysis
- Multivariate analysis

Univariate Analysis

- Univariate analysis is the simplest form of analyzing data.
- "Uni" means "one", so in other words, data has only one variable.
- It doesn't deal with causes or relationships and its major purpose is to describe. It takes data, summarizes that data, and finds patterns in the data.
- Univariate Analysis can be done either on **numerical or categorical** features.

Let's explore the types of plot for univariate analysis

Histogram

- In the univariate analysis, we use histograms for analyzing and visualizing frequency distribution.
- Plotting histograms in Pandas are very easy and straightforward.
- A **histogram** represents the distribution of data by forming bins along with the range of the data and then drawing bars to show the number of observations that fall in each bin.
- The bin can be of any size.

`Note: When dealing with a set of data, usually the first thing is to get a sense of how the variables are distributed.`
`We start by identifying a few variables of interest and checking their distribution.`

In order to make any prediction we need to fit a linear regression model, so we have to make sure the distribution of the variables is almost linear.

Now to check the linearity of the variables for any skewness in the distribution and outliers in the data.

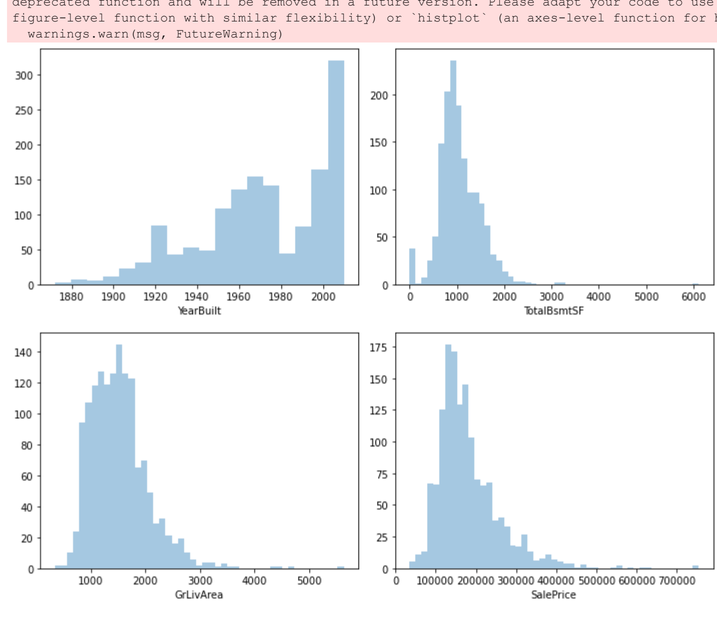
The variables we check for are 'OverallQual', 'YearBuilt', 'TotalBsmtSF', 'GrLivArea', 'Neighborhood' and the target variable 'SalePrice'.

Plot Histogram for 'YearBuilt', 'TotalBsmtSF', 'GrLivArea', and 'SalePrice'

In [11]:

```
import seaborn as sns

num_cols = ['YearBuilt', 'TotalBsmtSF', 'GrLivArea', 'SalePrice']

for i in range(0,len(num_cols),2):
        plt.figure(figsize=(10,4))
        plt.subplot(121)
        sns.distplot(dataframe[num_cols[i]], kde=False)
        plt.subplot(122)
        sns.distplot(dataframe[num_cols[i+1]], kde=False)
        plt.tight_layout()
        plt.show()
```

```
C:\Users\alpika.gupta\Anaconda3\lib\site-packages\seaborn\distributions.py:2619: FutureWarning: `distplot` is a deprecated function and will be removed in a future version. Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).
  warnings.warn(msg, FutureWarning)
```



Observations:

In the above graphs, the YearBuilt graph is left skewed and all the other graphs are right skewed. We need to normalize the data in EDA.

The Kernel Density Estimation

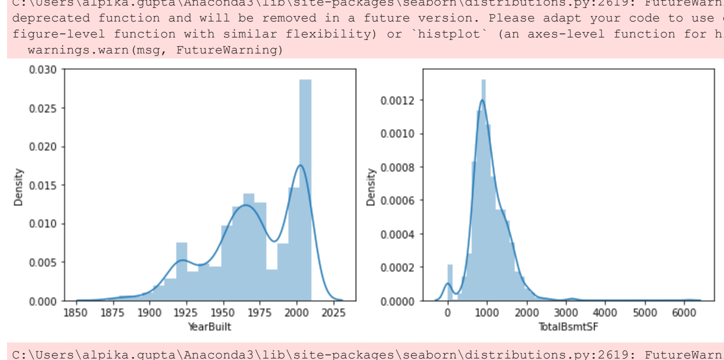
- it is also a useful tool for plotting the shape of a distribution.
- Like the histogram, the KDE plot encodes the density of observations on one axis with height along the other axis.

Plot Kernel Density Function for 'YearBuilt', 'TotalBsmtSF', 'GrLivArea', and 'SalePrice'

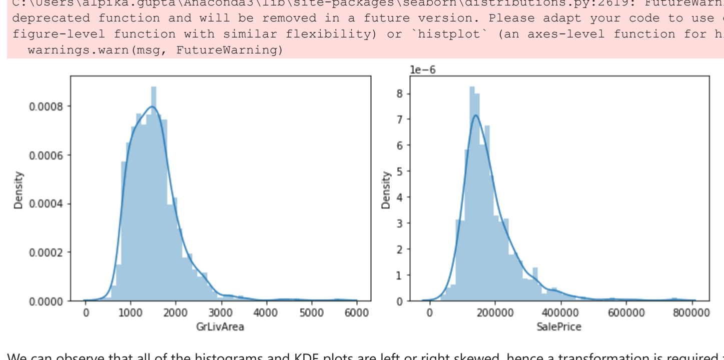
In [12]:

```
num_cols = ['YearBuilt', 'TotalBsmtSF', 'GrLivArea', 'SalePrice']

for i in range(0,len(num_cols),2):
        plt.figure(figsize=(10,4))
        plt.subplot(121)
        sns.distplot(dataframe[num_cols[i]], hist=True, kde=True)
        plt.subplot(122)
        sns.distplot(dataframe[num_cols[i+1]], hist=True, kde=True)
        plt.tight_layout()
        plt.show()
```

```
C:\Users\alpika.gupta\Anaconda3\lib\site-packages\seaborn\distributions.py:2619: FutureWarning: `distplot` is a deprecated function and will be removed in a future version. Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).
  warnings.warn(msg, FutureWarning)
```



```
C:\Users\alpika.gupta\Anaconda3\lib\site-packages\seaborn\distributions.py:2619: FutureWarning: `distplot` is a deprecated function and will be removed in a future version. Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).
  warnings.warn(msg, FutureWarning)
C:\Users\alpika.gupta\Anaconda3\lib\site-packages\seaborn\distributions.py:2619: FutureWarning: `distplot` is a deprecated function and will be removed in a future version. Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).
  warnings.warn(msg, FutureWarning)
```



We can observe that all of the histograms and KDE plots are left or right skewed, hence a transformation is required to make them linear.

Boxplots

- A box plot (or box-and-whisker plot) shows the distribution of quantitative data in a way that facilitates comparisons between variables or across levels of a categorical variable.
- The box shows the quartile of the dataset while the whiskers extend to show the rest of the distribution.
- Boxplots help to find the features which can be removed.
- This plot is also used in multivariate analysis.
- The box plot (a.k.a. box and whisker diagram) is a standardized way of displaying the distribution of data based on the five-number summary:
  - Minimum
  - First quartile
  - Median
  - Third quartile
  - Maximum
- In the simplest box plot, the central rectangle spans the first quartile to the third quartile (the interquartile range or IQR).
- A segment inside the rectangle shows the median and "whiskers" above and below the box show the locations of the minimum and maximum.

Outliers are either 3×IQR or more above the third quartile or 3×IQR or more below the first quartile.

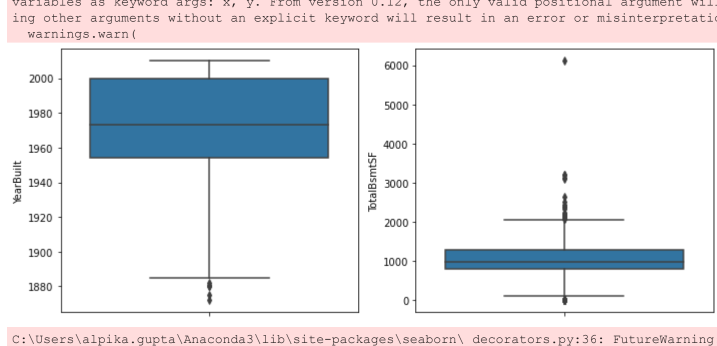
Suspected outliers are slightly more central versions of outliers, either 1.5×IQR or more above the third quartile or 1.5×IQR or more below the first quartile.

Plot Boxplots for 'YearBuilt', 'TotalBsmtSF', 'GrLivArea', and 'SalePrice'

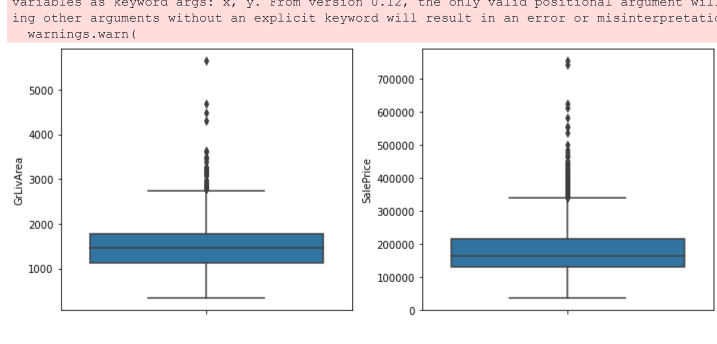
In [13]:

```
num_cols = ['YearBuilt', 'TotalBsmtSF', 'GrLivArea', 'SalePrice']

facet = None

for i in range(0,len(num_cols),2):
        plt.figure(figsize=(10,4))
        plt.subplot(121)
        sns.boxplot(facet, num_cols[i],data = dataframe)
        plt.subplot(122)
        sns.boxplot(facet, num_cols[i+1],data = dataframe)
        plt.tight_layout()
        plt.show()
```

```
C:\Users\alpika.gupta\Anaconda3\lib\site-packages\seaborn\_decorators.py:36: FutureWarning: Pass the following variables as keyword args: x, y. From version 0.12, the only valid positional argument will be `data`, and passing other arguments without an explicit keyword will result in an error or misinterpretation.
  warnings.warn(
C:\Users\alpika.gupta\Anaconda3\lib\site-packages\seaborn\_decorators.py:36: FutureWarning: Pass the following variables as keyword args: x, y. From version 0.12, the only valid positional argument will be `data`, and passing other arguments without an explicit keyword will result in an error or misinterpretation.
  warnings.warn(
```



```
C:\Users\alpika.gupta\Anaconda3\lib\site-packages\seaborn\_decorators.py:36: FutureWarning: Pass the following variables as keyword args: x, y. From version 0.12, the only valid positional argument will be `data`, and passing other arguments without an explicit keyword will result in an error or misinterpretation.
  warnings.warn(
C:\Users\alpika.gupta\Anaconda3\lib\site-packages\seaborn\_decorators.py:36: FutureWarning: Pass the following variables as keyword args: x, y. From version 0.12, the only valid positional argument will be `data`, and passing other arguments without an explicit keyword will result in an error or misinterpretation.
  warnings.warn(
```



Observation:

Plot Boxplots for 'YearBuilt', 'TotalBsmtSF', 'GrLivArea', and 'SalePrice' show some outliers, which are represented by dots.

Univariate Analysis for Categorical Variable

Most of the buildings are built post 1960 and most houses have an area in the range 1-2k sq feet. There are quite a few outliers in the sale price and living area and these might be correlated. We need to check the distribution of the categorical columns.

We chose the following columns from the above categorical_features_columns :

- Neighborhood
- SaleCondition

Countplot for Categorical Feature 'SaleCondition'

- This plots histograms of the variable

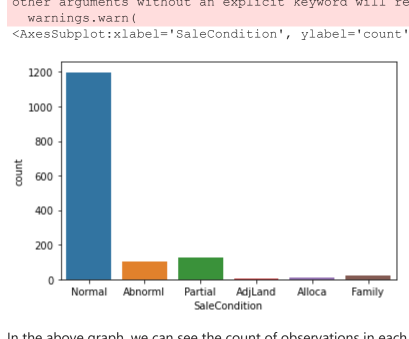
In [14]:

```
sns.countplot('SaleCondition', data=dataframe)
```

```
C:\Users\alpika.gupta\Anaconda3\lib\site-packages\seaborn\_decorators.py:36: FutureWarning: Pass the following variable as a keyword arg: x. From version 0.12, the only valid positional argument will be `data`, and passing other arguments without an explicit keyword will result in an error or misinterpretation.
  warnings.warn(
```

Out[14]:

```
<AxesSubplot:xlabel='SaleCondition', ylabel='count'>
```



In the above graph, we can see the count of observations in each categorical bin using bars.

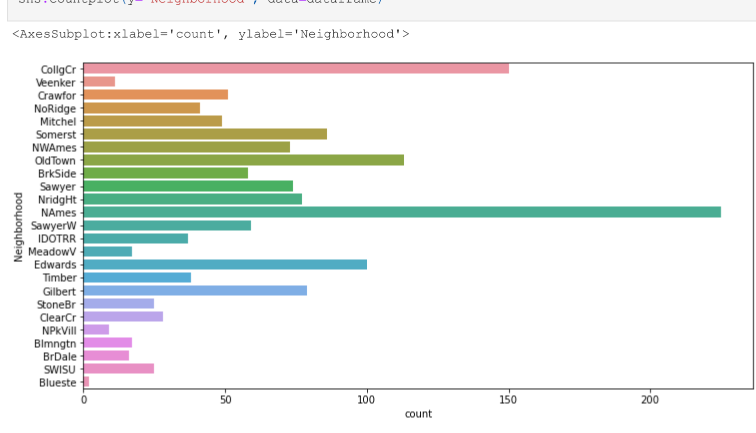
Countplot For Categorical Feature 'Neighborhood'

In [15]:

```
plt.figure(figsize=(12,6))

sns.countplot(y='Neighborhood', data=dataframe)
```

Out[15]:

```
<AxesSubplot:xlabel='count', ylabel='Neighborhood'>
```



In the above graph, we can see the count of observations in each categorical bin using bars in the y-axis.

Now we will plot some important feature against our target.

Multivariate Analysis

In multivariate analysis, we try to find the relations between multiple variables. Obviously, in real-life problems, variables can be any combination of numeric or categorical variables. The combinations are:

- Numeric vs. Numeric
- Numeric vs. Categorical
- Categorical vs. Categorical

Another aspect of variable combination are:

- Feature vs. Feature
- Feature vs. Target

Numeric vs. Numeric

For these kind of plots we use a scatterplot of the two variables. Although one can use a variety of plots in seaborn to do a quick and dirty EDA to some sophisticated plots. lmplot is one of the plots to do a scatterplot in seaborn, it'll by default fit a regression line on top which you can control using 'fit_reg' argument.

Now we will plot some important feature against our target.

Plot 'GrLivArea' against 'SalePrice'

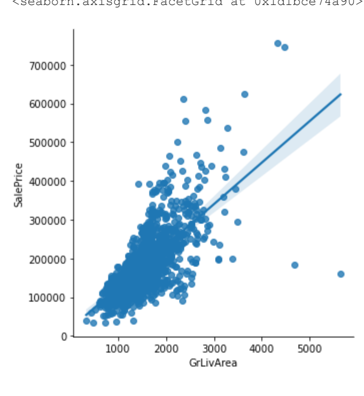
In [16]:

```
sns.lmplot('GrLivArea', 'SalePrice', data=dataframe, fit_reg=True)
```

```
C:\Users\alpika.gupta\Anaconda3\lib\site-packages\seaborn\_decorators.py:36: FutureWarning: Pass the following variables as keyword args: x, y. From version 0.12, the only valid positional argument will be `data`, and passing other arguments without an explicit keyword will result in an error or misinterpretation.
  warnings.warn(
```

Out[16]:

```
<seaborn.axisgrid.FacetGrid at 0x1d1bce74a90>
```



- Using a scatterplot we can also detect *multivariate outliers*, in this case there are two houses which have an area above ~4500 and they don't follow the trend. Removing these would give a better fit.

Another plot that we can use is a jointplot which gives a plethora of information in a single plot. It has:

- Scatter Plot
- Regression line fit to the data.
- Histogram and KDE of individual variables.
- Pearson correlation and p value.

Plot 'TotalBsmtSF' against 'SalePrice'

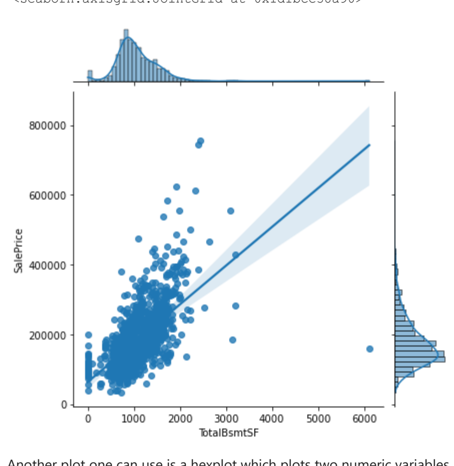
In [17]:

```
sns.jointplot('TotalBsmtSF', 'SalePrice', data=dataframe, kind='reg')
```

```
C:\Users\alpika.gupta\Anaconda3\lib\site-packages\seaborn\_decorators.py:36: FutureWarning: Pass the following variables as keyword args: x, y. From version 0.12, the only valid positional argument will be `data`, and passing other arguments without an explicit keyword will result in an error or misinterpretation.
  warnings.warn(
```

Out[17]:

```
<seaborn.axisgrid.JointGrid at 0x1d1bce38a90>
```



Another plot one can use is a hexplot which plots two numeric variables. Darker colors signify more points

Plot 'YearBuilt' against 'SalePrice'

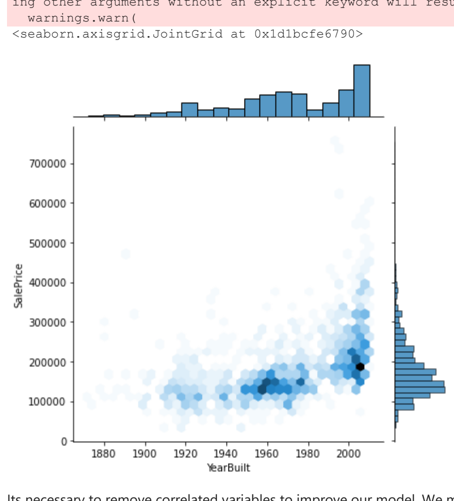
In [18]:

```
sns.jointplot('YearBuilt', 'SalePrice', data=dataframe, kind='hex')
```

```
C:\Users\alpika.gupta\Anaconda3\lib\site-packages\seaborn\_decorators.py:36: FutureWarning: Pass the following variables as keyword args: x, y. From version 0.12, the only valid positional argument will be `data`, and passing other arguments without an explicit keyword will result in an error or misinterpretation.
  warnings.warn(
```

Out[18]:

```
<seaborn.axisgrid.JointGrid at 0x1d1bcfe6790>
```



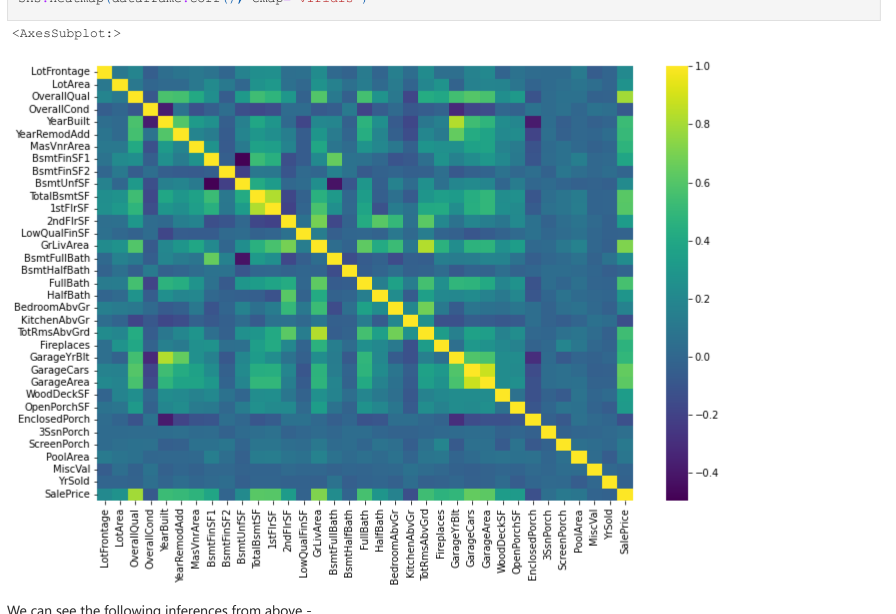
Its necessary to remove correlated variables to improve our model. We must find correlations can visualize the correlation matrix using a heatmap.

In [19]:

```
plt.figure(figsize=(12,8))
sns.heatmap(dataframe.corr(), cmap='viridis')
```

Out[19]:

```
<AxesSubplot:>
```



We can see the following inferences from above -

- A lot of variables are correlated to SalePrice which is good.
- GrLivArea is highly correlated with TotRmsAbvGrd.
- Also year the garage was built(GarageYrBlt) is correlated with the year the building was built(YearBuilt).

It's a good idea to remove such correlated variables during feature selection.

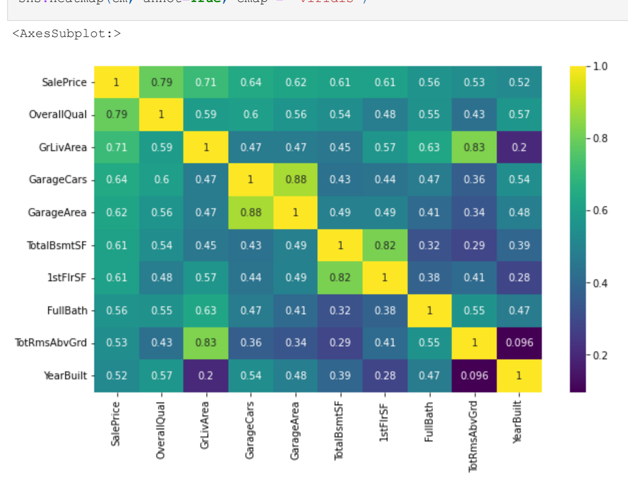
Now we plot some top variables to see the relations between them. We starts by filtering top 10 variables which are highly correlated with SalePrice.

In [20]:

```
#saleprice correlation matrix
k = 10 #number of variables for heatmap
cols = dataframe.corr().nlargest(k, 'SalePrice')['SalePrice'].index
cm = dataframe[cols].corr()
plt.figure(figsize=(10,6))
sns.heatmap(cm, annot=True, cmap = 'viridis')
```

Out[20]:

```
<AxesSubplot:>
```



- 'OverallQual', 'GrLivArea' and 'TotalBsmtSF' are strongly correlated with 'SalePrice'.
- 'GarageCars' and 'GarageArea' are also correlated variables. However no. of cars that can fit into a garage is dependent on the garage area and one can remove one of these. Check the correlation between them.
- TotalBsmtSF and 1stFlrSF are also highy correlated. We can drop one of these.
- As pointed out above Yearbuilt and TotRmsAbvGrd are highly correlated, we'll discard TotRmsAbvGrd.

An easy way to plot all possible interactions is between a set of numeric variables is using pairplot function in seaborn.

In [21]:

```
# Visualizing relations between all major variables
cols = ['SalePrice', 'OverallQual', 'GrLivArea', 'GarageCars', 'TotalBsmtSF', 'FullBath', 'YearBuilt']
sns.pairplot(dataframe[cols])
```

Out[21]:

```
<seaborn.axisgrid.PairGrid at 0x1d1bcd37610>
```